How Stable Are Regression Conclusions Across Subpopulations?

Nadjib Tebbal – A Finite-Sample Empirical Study

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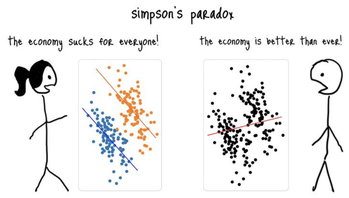
Introduction

The Problem

You fit a regression on the full dataset. R^2 is fine, p-values are small. A reviewer asks: *"Does this hold for each subgroup?"*

You check. It doesn't. Now what?

Simpson's Paradox



Within each group the trend is positive. Pooled, it reverses. The pooled regression hides subgroup structure.

"It's an Artifact"

"The coefs differ across groups!"

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"Genuine heterogeneity. Model doesn't transport."

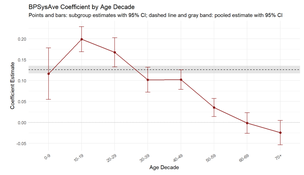
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"Wait — different X distributions cause most of the drift."

Coef differences = **genuine heterogeneity** + **mechanical artifact** (covariate distribution differences). Standard practice conflates them.

Real-Data: NHANES Blood Pressure

$n = 9{,}254$, systolic BP, age-decade subgroups.

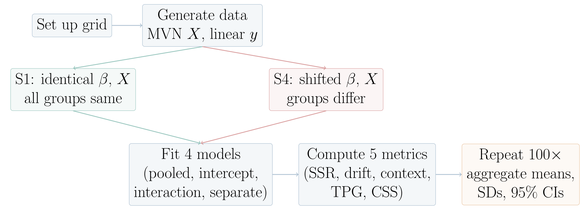


SSR = 0.57 | **Drift: 32.07 → 7.49** (context-norm.)

- Effect *reverses sign* in older groups
- Pooled model underperforms in **every** age decade
- $\widetilde{D}_j$ stays large after correction

The Work: Simulation Study

How the Simulation Works



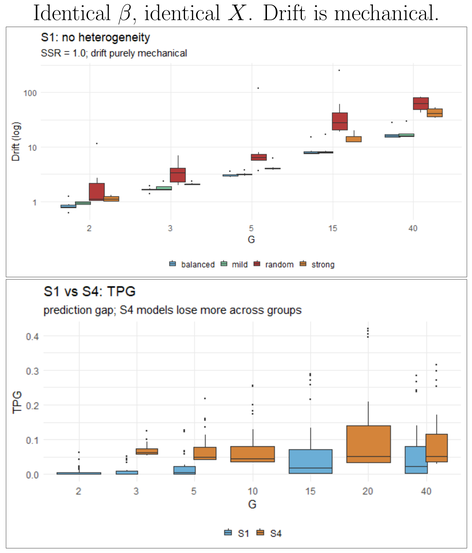
Five Stability Metrics

Metric	Definition	What it captures
SSR	$\mathrm{SSR}_j = \frac{1}{M}\sum \mathbf{1}(\operatorname{sgn}\hat{\beta}_{j,a} = \operatorname{sgn}\hat{\beta}_{j,b})$	Do signs $(+/-)$ reverse across groups?
Std. drift	$D_j^\star = \frac{\max_g \lvert\hat{\beta}_{j,g} - \hat{\beta}_{j,\text{pooled}}\rvert}{\widehat{\mathrm{SE}}(\hat{\beta}_{j,\text{pooled}})}$	How far do subgroup coefs move from pooled?
Context drift	$\widetilde{D}_j = \frac{D_j^\star}{1+M_j}, \quad M_j = \text{avg. } \lvert\bar{x}_{j,a} - \bar{x}_{j,b}\rvert / s_j$	Drift corrected for X distribution differences.
TPG	$\mathrm{TPG}_{a\to b} = \mathrm{RMSE}_{\text{pooled}\to b} - \mathrm{RMSE}_{b\to b}$	Prediction loss across groups.
CSS	$\mathrm{CSS}_{a,b} = \frac{1}{p}\sum_j \frac{\lvert\bar{x}_{j,a} - \bar{x}_{j,b}\rvert}{s_j}$	How different are X distributions?

Simulation Results: S1 vs S4

Scenario 1: No Heterogeneity

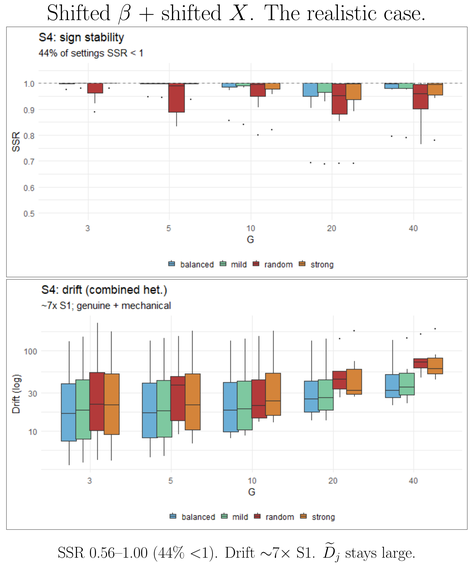
Identical β, identical X. Drift is mechanical.



SSR = 1.0 throughout. Context shrinks drift ∼35%.

Scenario 4: Combined Heterogeneity

Shifted β + shifted X. The realistic case.



SSR 0.56–1.00 (44% <1). Drift ∼7× S1. $\widetilde{D}_j$ stays large.

Key contrast: S1 drift is purely mechanical ⇒ $\widetilde{D}_j \approx 0$ after correction. S4 drift is mechanical + genuine ⇒ $\widetilde{D}_j$ stays large. The three-regime diagnostic works as predicted.

Why it Happens: The Explanation

Key Idea: $D_j^\star = \widetilde{D}_j \times (1 + M_j)$

Different X distributions ⇒ different design matrices ⇒ mechanical drift even with identical β.

$$D_j^\star \;=\; \widetilde{D}_j \;\times\; (1 + M_j)$$

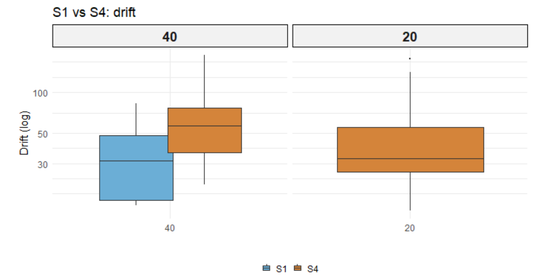
Total = Genuine × Mechanical

$$M_j \approx 0 \;\Rightarrow\; \widetilde{D}_j \approx D_j^\star \text{ (real)}$$

$M_j \uparrow,\ \widetilde{D}_j \downarrow \Rightarrow$ artifact

$M_j \uparrow,\ \widetilde{D}_j \uparrow \Rightarrow$ genuine

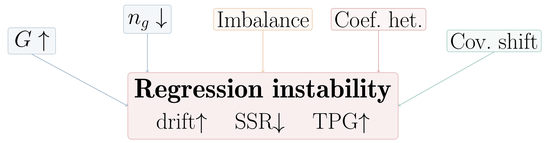
S1 vs S4: What Differed?



S4 breaks: coefficient shifts cause sign reversals (SSR < 1); covariate shifts add mechanical drift. Context strips the mechanical part but β differences remain.

S1 stable: no true heterogeneity ⇒ all drift mechanical ⇒ $\widetilde{D}_j \to 0$.

What Drives Instability?



- These factors interact multiplicatively.
- Always check per-group n and imbalance before interpreting subgroup estimates.

References & Future Work

Future: Run S2–S3; then collinearity, higher p, different models.

References:

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